JewlX Platform

Raat ki 3 Bhari Jobs: Ye Asal Mein Kya Karti Hain?

Backup, Diamond pipeline aur Availability refresh - non-technical explainer, taake faisla kiya ja sake ke kis ko rokna, kis ko weekly karna aur kis ko sirf halka karna hai.

Pehle ek tasveer zehn mein rakhein

JewlX ka database ek bara **godam** hai. Is godam ka sab se bara hissa **diamond pool** hai: takreeban **7 lakh 94 hazar heeray** (794,000 rows), jo aap CSV sheets se upload karte hain. Baqi sab kuch (clients, leads, orders, settings) chhota hai, chand hazar rows.

Raat ko 3 'staff members' is godam mein kaam karte hain. Teeno ka kaam zaroori lagta hai, lekin teeno **poore godam ka chakkar** lagate hain, aur yahi DB par load ki asal wajah hai. Neeche har ek ka kaam, maqsad, aur 'agar rok diya to kya hoga' aasan alfaz mein hai.

Job	Aasan naam	Kab chalti hai (PKT)	Ek line mein kaam
daily-jewlx-db-backup	Raat ka backup	Roz raat 01:00	Poore database ki ek copy file bana kar Storage mein rakhna
diamond-pipeline-tick + diamond-pipeline-nightly	Diamond audit	Chowkidar har minute; poora audit roz subah 08:07 aur har upload ke baad	Har heeray ka price check, har group ka 'best heera' chunna, filter lists banana
refresh_stale_diamond_availability_5min	Availability list	Har 5 minute (din-raat)	Har client ke liye 'kaunsi shape/carat stock mein hai' ki list dobara banana

1. Raat ka Backup (daily-jewlx-db-backup)

Ye kya karti hai?

Roz raat 01:00 PKT (20:00 UTC) ek program chalta hai jo database ki **har table** ki poori copy utha kar ek bari JSON file banata hai, use zip karta hai, aur Supabase Storage ke **jewlx-backups** folder mein rakh deta hai. 30 din se purani files khud delete ho jati hain. Admin panel ke **Data Governance** page par in backups ki list dikhti hai, wahan se download bhi ho sakti hain aur 'Run backup' ka button bhi hai.

Iska maqsad kya hai?

Insurance. Agar kabhi koi galti se data delete kar de, ya koi bug tables kharab kar de, to kal raat wali copy se data wapas laya ja sake.

Ye bhari kyun hai?

Masla ye hai ke ye program **chhote aur bare tables mein farq nahi karta**. Clients, leads, orders jaisi tables chand hazar rows ki hain; wo seconds mein copy ho jati hain. Lekin diamond pool (794,000 rows) aur uske saath scores wali table (phir 794,000 rows) bhi poori copy hoti hain, yani **16 lakh rows roz raat**.

Aur ye copy 1000-1000 rows ke tukron mein hoti hai. Har tukre ke liye database ko kaha jata hai 'page 1 se shuru karo aur 500,000 rows chhor kar agli 1000 do'. Ye aisa hai jaise kisi kitaab ka page 700 kholne ke liye har baar page 1 se ginna shuru karein. Jitna aage jayen, utna waqt. Is liye ye job **1 se 2 ghante** database ko mast rakhti hai, aur 24 October ko alert isi ke doran (shuru hone ke 2 ghante baad) aaya.

Ek aur baat: diamond pool ka data to aapki apni CSV sheets hain. Agar kabhi kho bhi jaye to dobara upload ho sakta hai. Is ko roz backup karna mehnat zyada, faida kam.

Agar isay weekly kar dein to kya hoga?

- **Asal risk:** is copy mein 7 din tak ka data na hone ka imkaan (agar beech mein kuch kharab ho). Lekin Supabase khud bhi (Pro plan par) roz apna backup leta hai jo dashboard ke Database > Backups mein dikhta hai. Pehle ye confirm kar lein ke wahan daily backups nazar aa rahe hain; agar haan, to JewlX wala backup sirf 'doosri copy' hai aur weekly bilkul kaafi hai.
- **Load:** raat ka 1-2 ghante wala spike hafte mein 6 raatein khatam.
- **Behtar tajweez:** weekly bhi karein AUR diamond pool/scores tables ko backup se nikaal dein. Phir ye job chand seconds ki reh jayegi, chahe weekly ho ya daily. Aap chahen to chhoti tables (clients, leads, orders) roz aur poora weekly, dono rakh sakte hain.

Meri raye: Weekly (itwaar subah 09:30 PKT, jab diamond audit khatam ho chuka ho) + diamond tables exclude. Ye sab se kam risk aur sab se zyada faida wala option hai.

2. Diamond Audit (diamond-pipeline-tick, diamond-pipeline-nightly)

Ye 2 cron entries hain lekin asal mein **ek hi kaam** ke do hisse:

Entry	Kirdar	Kitna bhari
diamond-pipeline-tick (har minute)	**Chowkidar.** Har minute sirf ek chhoti si parchi dekhta hai: 'kya godam mein naya maal aaya hai?' Agar nahi, 2 millisecond mein wapas. Agar haan (upload ke 2 minute baad), to poora audit shuru karta hai.	Idle: kuch nahi. Audit: 10-60 min
diamond-pipeline-nightly (roz 08:05 PKT)	**Zabardasti ki parchi.** Roz subah parchi par 'naya maal aaya' likh deta hai, chahe aaya ho ya nahi. 2 minute baad chowkidar poora audit chala deta hai.	Khud kuch nahi, lekin roz audit karwata hai
Upload trigger (cron nahi)	Jab bhi aap sheet upload/append/merge karte hain, parchi khud 'naya maal' ho jati hai.	Har upload = ek audit

Poora audit kya karta hai?

Jab audit chalta hai to har ek heeray ko haath lagaya jata hai, aur 4 cheezein tayyar hoti hain:

• **Price check (outlier scoring):** har heeray ka price uske jaise doosre heeron (same shape, carat, color, clarity) se compare hota hai. Jo heera shak ki had tak sasta ho (galat price, typo, ya mashkook listing) use 'hide' ka flag milta hai. Ye wo **Diamond Outlier engine** hai jo abhi **shadow mode** mein hai: scores ban rahe hain lekin sirf **Sandbox client** par asal mein heere chhupaye ja rahe hain, baqi clients par sirf report.

• **Har group ka 'best heera' (group representatives):** Ring OS widget ke Stone step par customer ko har shape + carat + color + clarity ke liye **ek** heera dikhaya jata hai, 794,000 nahi. Ye audit har group ka sab se munasib heera chun kar ek chhoti list (materialized view) mein rakhta hai. **Ring OS aur Quote Estimator dono isi list se stones dikhate hain, har client ke liye.**

• **Grade stats:** har group mein kitne heere hain, price range kya hai (Catalog page ke counts).

• **Filter lists:** Catalog page ke dropdowns (shapes, colours, clarities, currencies) ki values. Is ke liye poora pool 9 baar alag alag ginna jata hai.

Ye bhari kyun hai?

Ek audit mein poora pool **15 se 18 baar** parha jata hai aur ~16 lakh rows likhi jati hain (scores table poori khali kar ke dobara bhari jati hai). 794,000 heeron par ye 10 se 60 minute ka kaam hai jis mein CPU aur disk dono 100% par hote hain. Code mein is job ko **1 ghante tak** chalne ki ijazat di gayi hai.

Aur ye roz hota hai, chahe pool mein ek bhi heera na badla ho. Upar se jis din aap 3 sheets upload karein, us din 3 audit aur chalte hain.

Agar rok dein to kya hoga?

• **Chowkidar (tick) ko mat rokein.** Agar ye band ho gaya to upload ke baad naye heere Ring OS/Quote Estimator mein **kabhi nahi dikhenge**, kyun ke 'best heera' list purani rahegi. Ye idle halat mein kuch kharch bhi nahi karta.

• **Nightly wali zabardasti parchi (08:05) ko rokna bilkul safe hai.** Ye sirf 'safety net' thi: agar kabhi upload wala audit fail ho jaye to raat ko dobara ho jaye. Is ki jagah 2 behtar options hain: (a) raat ko sirf tab chalao jab pool waqai badla ho aur aakhri audit uske baad na hua ho; ya (b) hafte mein ek baar (itwaar) safety net.

• **Ek risk jo samajhna zaroori hai:** agar audit beech mein rok diya jaye (jaise Lovable ka 'time limit' wala suggestion) to scores table khali ho kar reh sakti hai, aur Sandbox par heere ghalat chhup sakte hain. Is liye audit ko **kam baar** chalana theek hai, **adhoora chhorna** nahi.

Meri raye: Nightly ko 'sirf agar pool badla ho' bana dein (quiet raaton ko kuch nahi chalega), ya weekly kar dein. Tick wahi rahe. Saath mein debounce 2 se 10 minute kar dein taake ek session mein 3 sheets upload karne par 1 audit ho, 3 nahi.

3. Availability List (refresh_stale_diamond_availability_5min)

Ye kya karti hai?

Har client apne admin panel mein **Catalog** page par filter set karta hai: 'mere widget mein sirf ye shapes, ye carat range, natural ya lab dikhao', phir **Confirm** dabata hai. Us ke baad Ring OS widget ke Shape aur Carat steps par customer ko sirf wo options roshan dikhte hain jo us client ke filter ke andar **asal mein stock mein hain**; baqi grey ho jate hain.

'Kya stock mein hai' ka jawab nikalne ke liye poore pool ko client ke filter se chhanna parta hai. Ye mehenga hai, is liye jawab ek **cache** (tayyar-shuda parchi) mein rakha jata hai, aur widget sirf parchi parhta hai. Ye job har 5 minute dekhti hai ke kis client ki parchi **purani (stale)** ho gayi hai aur 5 clients tak ki parchi dobara banati hai.

Iska maqsad kya hai?

Confirm Catalog ke baad widget mein naye filters jaldi reflect hon, bina customer ko slow query ka intezaar karwaye. Ye theek maqsad hai aur isay rakhna chahiye.

Ye bhari kyun hai? (2 wajuhaat)

- **24 ghante wala rule:** code kehta hai parchi 'purani' tab bhi hai jab wo 24 ghante se zyada pehle bani ho, chahe client ne kuch na badla ho. Natija: **har confirmed client ki parchi roz dobara banti hai**, din bhar 5-5 kar ke. Jitne clients, utne poore pool ke chakkar roz.
- **Chhanne ka tareeqa:** filter ek chhoti si program (function) ke zariye **har heeray par alag alag** lagaya jata hai, is liye database apne indexes (fehrist) use nahi kar pata aur hamesha poora godam ghoomta hai. Ek client ki parchi banane mein 5 se 40 second poora CPU lagta hai.

Ek aur ahem baat: ye job **repo (code) mein mojood nahi**. September mein ise code se hata diya gaya tha lekin ye database mein pehle hi laga chuki thi, is liye wahan ab tak chal rahi hai. Yahi wajah hai ke Lovable aur code review dono ise nahi dekhte.

Agar rok dein to kya hoga?

- **Poori tarah rokna theek nahi.** Repo mein Confirm Catalog par parchi dobara banane ka koi aur raasta nahi (wo code bhi September mein revert ho gaya tha). Job band hui to Confirm ke baad widget **purani availability** dikhata rahega.
- **Sahi fix:** 24-ghante wala rule hata dein. Parchi sirf tab bane jab client ne waqai Confirm kiya ho (stale flag) ya naya pool upload hua ho. Phir ye job din mein shayad 2-4 baar asal kaam karegi, 288 baar nahi.
- **Doosra fix (saath mein):** chhanne ka tareeqa badal kar seedha database columns par filter lagayein taake indexes use hon. Natija bilkul wahi, waqt 10 guna kam.
- Aur is job ko **repo mein wapas likhein** taake code aur database ek jaise hon.

Meri raye: Rokna nahi, halka karna: 24h rule hatao + set-based filter. Load 90% kam, widget ka behaviour same.

4. Ek Nazar Mein: Rokein, Weekly Karein, ya Halka Karein?

Job	Asal maqsad	Agar poori tarah band ki	Tajweez	Load bachat
Raat ka backup	Database ki insurance copy, admin se download	Sirf JewlX wali doosri copy band hogi; Supabase ka apna backup chalta rahega (dashboard mein confirm karein)	Weekly (itwaar) + diamond pool/scores exclude	~25-35% total, raat ka 2-ghante spike khatam
Diamond audit - nightly flag	Safety net: roz zabardasti poora audit	Kuch nahi, jab tak upload wala audit chalta rahe	Band karein ya 'sirf agar pool badla' / weekly	~30-40% on quiet days
Diamond audit - tick (har minute)	Upload ke baad naye heere widgets mein lana	Naye heere kabhi nahi dikhenge - MAT rokein	Rakhein; debounce 2 -> 10 min	Idle par koi load nahi
Availability list (5 min)	Confirm Catalog ke baad widget mein sahi shapes/carats	Widget purani availability dikhayega	Rakhein lekin 24h rule hatao + tez filter	~10-20%, din bhar ka hum khatam

Aap ke faisle ke liye 3 sawal

• **Backup:** Weekly theek hai? Aur kya diamond pool/scores tables ko backup se nikaal dein (wo CSV se dobara upload ho sakti hain)? Pehle Supabase dashboard > Database > Backups mein dekh lein ke platform ke apne daily backups mojood hain.

• **Diamond nightly audit:** 'sirf jab pool badla ho' (recommended) ya 'hafte mein ek baar'? Dono mein tick (chowkidar) chalta rahega.

• **Availability:** 24h rule hatane par theek hain? (Confirm Catalog ke baad behaviour bilkul same rahega, sirf faltu roz ka kaam band hoga.)

Jab aap in 3 par haan/na bata dein, to agla qadam 3-4 chhoti migrations hain (koi widget, pricing ya lead logic nahi badalta). Un ke 3-4 din baad Supabase ke CPU graph dekh kar Medium instance par wapas jaya ja sakta hai (~210 credits/month bachat).

Technical references (agar zaroorat ho)

Job	Code
Backup	supabase/functions/database-backup/index.ts; cron: supabase/migrations/20260421135203_*.sql; UI: src/pages/admin/DataGovernance.tsx; bucket jewlx-backups, 30-day retention
Diamond audit	diamond_pipeline_tick + score_diamond_pool: 20260925090000_diamond_outlier_engine.sql, 20261022120000_ring_os_catalog_currency_filter.sql, 20261025120000_diamond_pool_distinct_values_cache.sql; widgets read via get_diamond_presets_filtered / get_quote_estimator_diamonds
Availability	refresh_stale_diamond_availability(5): 20260730195243_*.sql; compute_client_diamond_catalog_availability: 20260910124634_*.sql; cron file deleted from repo (git show ff79772d:supabase/migrations/20260910130515_*.sql)